Inorganic Cume
January 16, 2004

bond length of 2.14Å.

The anion $Mo_2Cl_8^{-4}$ is known to have a Mo-Mo bximately at right angles The four chlorine atoms on each Mo are bonded appr try. It may be prepared to the Mo-Mo bond axis and are in an eclipsed geome according to the following reactions:

$$Mo(CO)_6 \xrightarrow[\text{Acetic Acid}]{\text{diglyme}} Mo_2(O_2CCH_3)_4 \xrightarrow[0^oC]{HCl(\text{aq.})} Mo_2Cl_8^{-4}$$

eep in mind that the

1) Describe the bonding in this molecule. K. molecule is diamagnetic.

2) Accou bond lengths observed in the nt for the difference in Mo following complexes.

$Mo_2(SO_4)_4$ $)_4^{-3}$, 2.16Å; $Mo_2(HPO_4)_4^{-2}$, 2.23Å $^{-4}$, 2.11Å; $Mo_2(SO_4$

3) How would you about determining whether or not a Mo-Mo bond existed in have gone molecule if you had been the first one to prepare it. (2 this more ways which need not be completely independent.)